T17 Sandbox: Native vs Docker

Technical Advanced

Choosing the right isolation level for your context

Compare the actual boundaries

Claude Code's native Bash sandbox uses OS process isolation. Docker Sandboxes provides a separate environment for an agent, with local microVM and cloud options. The current Docker documentation uses the `sbx` CLI; check its installation and platform requirements before adopting commands from older `docker sandbox` examples.

Aspect	Native Claude sandbox	Docker Sandboxes
Boundary	OS process isolation	Separate sandbox environment
Kernel	Shares the host kernel	Local microVM has a separate kernel
Setup	`/sandbox`; Linux requires dependencies	Install and configure `sbx`
Workspace	Current and explicitly added directories	Configured workspace mounts and synchronization
Network	Domain policy and permission flow	Sandbox network policy
Cost and performance	Measure your commands	Measure compute and model use separately

Native sandbox

macOS uses Seatbelt; Linux and WSL2 use bubblewrap and socat. Native Windows and WSL1 are not supported by Claude Code's native sandbox.

```
/sandbox
```

In regular permissions mode, existing permission rules still apply. Auto-allow reduces prompts for sandboxed commands, with documented exceptions. Inspect the unsandboxed retry policy before relying on isolation.

Docker Sandboxes

Follow the current Docker Sandboxes documentation for `sbx`, supported platforms, local/cloud differences, hardware access and workspace behavior. Do not assume an older Docker Desktop version or a historical platform table describes the installed release.

A microVM adds a kernel boundary but does not guarantee immunity from escape, misuse of credentials or damage to mounted workspaces. Bypassing agent permission prompts still requires deliberate restrictions on accessible data and external services.

Choosing an environment

- Use the native sandbox for commands whose required access fits its filesystem and network controls.
- Evaluate a separate sandbox when the workload needs stronger separation or its own services.
- For unattended work, verify credentials, mounts, network rules, budget and recovery behavior before granting autonomy.
- Measure overhead and completed-task outcomes instead of relying on universal CPU or memory percentages.

Open-source runtime

Anthropic also publishes a sandbox runtime that can wrap other commands:

```
npx @anthropic-ai/sandbox-runtime node mcp-server.js
```

Source: Claude Code sandboxing.